Program : **Diploma in Cloud Computing and Big Data**	
Course Code : **6277**	Course Title: **Image Processing Lab**
Semester : **6**	Credits: **2.5**
Course Category: **Programme core course**	
Periods per week: **4 (L: 0 T: 1 P: 3)**	Periods per semester: **60**

Course Objectives:

- To implement basic and advanced image processing algorithms
- Able to implement various image enhancement methods
- Able to implement different image augmentation techniques
- An exposure of Convolutional neural network based feature extraction for classification problems

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Learn programming essentials	2139	Problem Solving and Programming	II
Develop python programs using Functions	3279	Programming python Lab	III
Concept of Machine Learning Techniques		Machine Learning	IV
Machine Learning Algorithms		Machine Learning Lab	V

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Practice various image processing libraries and demonstrate basic operations on image	10	Applying

CO2	Use different data augmentations and denoising techniques for improve the image dataset	10	Applying
CO3	Implement image enhancement methods and image gradients	19	Applying
CO4	Apply Convolutional neural networks for image feature extraction and classification	15	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3		3			
CO2	3	3	2	3			
CO3	3	3	2	3			
CO4	3	3	2	3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Practice various image processing libraries and demonstrate basic operations on image**		
M1.01	Practice the libraries: - OpenCV, Pillow, Matplotlib , PIL	4	Applying
M1.02	Implement program to Change the color spaces, thresholding operations on image	3	Applying
M1.03	Compute the mean, Standard deviation, correlation coefficient of a given image	3	Applying
Contents:	Illustrations of python programming using Anaconda platform/google colaboratory/jupyter lab/jupyter notebook , Familiarization of OpenCV, Pillow, Matplotlib, and/or PIL libraries. Reading, displaying operations of image, RGB to gray conversion, Applying statistical operations on images.		

CO2	**Use different data augmentations and denoising techniques for improve the image dataset**		
M2.01	Implement blurring, flipping, resizing, rotation operations	6	Applying
M2.02	Practice different denoising methods: Averaging, median filtering	4	Applying
Contents:	Implementing image augmentations techniques like blurring, flipping, translations. Resizing on images. Denoising of images		
	Lab Test – I	3	Applying
CO3	**Implement image enhancement methods and image gradients**		
M3.01	Apply 2D-FFT on an image	3	Applying
M3.02	Demonstrate stretching of a low contrast image, Histogram, and Histogram Equalization	4	Applying
M3.03	Perform Image slicing operations on given image	4	Applying
M3.04	Implement of Morphological operations on image	4	Applying
M3.05	Practice Edge Detection using Gradient Filters and canny edge algorithm	4	Applying
Contents:	Performing Fourier transform on image. Implementing contrast and brightness enhancement operations to enhance image. Implementing various operators like Sobel, Prewit, canny on images.		
CO4	**Apply Convolutional neural networks for image feature extraction and classification**		
M4.01	Implement Image feature extraction using Convolutional neural network	3	Applying
M4.02	Practice Image classification using Convolutional neural network	3	Applying
M4.03	Experiment with pretrained models such as VGG, Google Net, ResNet or Xception	6	Applying
M4.04	Open ended experiment**	3	Applying

Contents:	Implement convolutional filtering methods. Feature extraction with kernel size 3x3, 5x5. Implementation Image classification algorithm on iris dataset. Handwriting recognition algorithm on MNIST dataset. **** - Sample Open Ended Experiments** (Not evaluated for End Semester Examination but to be included in Continuous Internal Evaluation.) Students can-do open-ended real time projects/ experiments as a group of 2-3. There is no duplication in experiments between groups. <u>Consider the following sample scenarios:</u> 1. Fingerprint recognition 2. Face recognition Students should gather the required information, develop algorithms, and implement it in Python languages.		
	Lab Test – II	3	

Text / Reference

T/R	Book Title/Author
T1	Rafael C.Gonzalez & Richard E.Woods – Digital Image Processing – Pearson Education- 2/e – 2004.
T2	Anil.K.Jain – Fundamentals of Digital Image Processing- Pearson Education- 2003.
R3	William K. Pratt – Digital Image Processing – John Wiley & Sons-2/e, 2004

Online Resources

Sl.No	Website Link
1	https://www.geeksforgeeks.org/image-processing/
2	https://www.analyticsvidhya.com/blog/2021/08/image-processing-in-python-the-computer-vision-techniques/
3	https://www.datacamp.com/courses/image-processing-in-python